



Université Marie et Louis Pasteur

Course: Security for Connected Objects (Master M2 IOT)

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Assignment Title:

Practical Work (Modular Arithmetic & Cryptography)

Exercise 1: Understanding the Modulus

1. Calculate C by hand

Formula: $C = (P + k) \pmod{m}$

Substitute the given values: $C = (20 + 7) \pmod{26}$

Addition step: $20 + 7 = 27$

Modulo step: To find $27 \pmod{26}$, divide 27 by 26 and find the remainder.

$$27 = (26 \times 1) + 1.$$

Final Result: $C = 1$, because 1 is remainder.

2. What does mod 26 actually do to the result?

It acts as a "wrap-around" mechanism. Whenever a calculated number reaches or exceeds 26, the modulo operation divides the number by 26 and outputs only the remainder. It confines any infinitely growing number into a strict, repeating cycle.

3. What possible set of values can C take?

Because you are dividing by 26 and looking at the remainder, the only possible remainders are integers from 0 up to one less than the modulus.

Set of values: $\{0, 1, 2, 3, \dots, 25\}$.

4. Why does m=26 make sense when working with the alphabet?

The standard English alphabet contains exactly 26 letters. By assigning each letter a number (A=0, B=1, ... Z=25), using modulo 26 ensures that if you shift past 'Z' (25), the count wraps perfectly back around to 'A' (0).

Is $m=26$ a secret key?

No, $m = 26$ is not a secret key. It is a public parameter defined by the environment (the alphabet). The only secret piece of information needed to secure the message is the shift value, $k = 7$.

Exercise 2: Modular Addition Encryption

1. Calculate all ciphertexts without using Python

- **Formula:** $C = (P + 7) \pmod{26}$
- For $P = 0$: $0 + 7 = 7 \Rightarrow 7 \pmod{26} = 7$
- For $P = 1$: $1 + 7 = 8 \Rightarrow 8 \pmod{26} = 8$
- For $P = 5$: $5 + 7 = 12 \Rightarrow 12 \pmod{26} = 12$
- For $P = 10$: $10 + 7 = 17 \Rightarrow 17 \pmod{26} = 17$
- For $P = 19$: $19 + 7 = 26 \Rightarrow 26 = (26 \times 1) + 0 \Rightarrow C = 0$
- For $P = 20$: $20 + 7 = 27 \Rightarrow 27 = (26 \times 1) + 1 \Rightarrow C = 1$
- For $P = 25$: $25 + 7 = 32 \Rightarrow 32 = (26 \times 1) + 6 \Rightarrow C = 6$

3. Compare manual calculations with Python The manual calculations yield $\{7, 8, 12, 17, 0, 1, 6\}$, which perfectly match the expected output of the Python script.

4. What happens mathematically when $P + k \geq 26$? The value resets. Mathematically, 26 is subtracted from the sum (or a multiple of 26 is subtracted) to map the result back into the valid alphabet range.

5. Why does the ciphertext always stay in the range 0 to 25? By definition, the modulo operation ($x \pmod{m}$) outputs the remainder of x divided by m . A remainder of division by 26 can never be less than 0 or greater than 25.

6. Reflection: Decryption requirements If someone knows the formula and $m = 26$, the only missing piece of information they need to read the original message is the secret key, $k = 7$.

Exercise 3: Discovering Decryption

1. Formula to undo encryption

If encryption applies addition $(+k)$, the mathematical operation to undo it is subtraction $(-k)$.

The decryption formula is: $P = (C - k) \pmod{m}$.

2. Decrypting C = 1 (with m=26, k=7)

Solution:

(a) By Hand:

$$P = (1 - 7) \pmod{26} \quad P = -6 \pmod{26}$$

In modular arithmetic, to find the positive remainder of a negative number, add the modulus until you enter the positive range: $-6 + 26 = 20$.

Result: $P = 20$.

Python verification:

`print((1 - 7) % 26)` and the outputs 20.

(b) Re-encrypt P = 20:

$$C = (20 + 7) \pmod{26} = 27 \pmod{26} = 1. \text{ The process perfectly reverses itself.}$$

(c) Why subtraction works:

Because $(P + k) - k = P + (k - k) = P + 0 = P$. The modulo operation preserves standard arithmetic balances.

Understanding Check The decryption key is not a completely different random number; it is strictly related to k . In an additive cipher, the decryption key is just the same key k used with a subtraction operation (or it can be mathematically viewed as adding the inverse: $26 - 7 = 19$).

Exercise 4: Modular Multiplication

1. Calculate C by hand

- **Formula:** $C = (P \times k) \pmod{m}$

- **Given:** $P = 20, k = 5, m = 26$
- **Multiplication step:** $20 \times 5 = 100$
- **Modulo step:** $100 \pmod{26}$
- $100 = (26 \times 3) + 22$
- **Final Result:** $C = 22$

2. Evaluate the encryption function for all $P \in \{1,2,3, \dots, 25\}$

Using $C = (P \times 5) \pmod{26}$:

- $P = 1 \Rightarrow (1 \times 5) \pmod{26} = 5$
- $P = 2 \Rightarrow (2 \times 5) \pmod{26} = 10$
- $P = 3 \Rightarrow (3 \times 5) \pmod{26} = 15$
- $P = 4 \Rightarrow (4 \times 5) \pmod{26} = 20$
- $P = 5 \Rightarrow (5 \times 5) \pmod{26} = 25$
- $P = 6 \Rightarrow (6 \times 5) \pmod{26} = 4$
- $P = 7 \Rightarrow (7 \times 5) \pmod{26} = 9$
- $P = 8 \Rightarrow (8 \times 5) \pmod{26} = 14$
- $P = 9 \Rightarrow (9 \times 5) \pmod{26} = 19$
- $P = 10 \Rightarrow (10 \times 5) \pmod{26} = 24$
- $P = 11 \Rightarrow (11 \times 5) \pmod{26} = 3$
- $P = 12 \Rightarrow (12 \times 5) \pmod{26} = 8$
- $P = 13 \Rightarrow (13 \times 5) \pmod{26} = 13$
- $P = 14 \Rightarrow (14 \times 5) \pmod{26} = 18$
- $P = 15 \Rightarrow (15 \times 5) \pmod{26} = 23$
- $P = 16 \Rightarrow (16 \times 5) \pmod{26} = 2$
- $P = 17 \Rightarrow (17 \times 5) \pmod{26} = 7$
- $P = 18 \Rightarrow (18 \times 5) \pmod{26} = 12$
- $P = 19 \Rightarrow (19 \times 5) \pmod{26} = 17$
- $P = 20 \Rightarrow (20 \times 5) \pmod{26} = 22$
- $P = 21 \Rightarrow (21 \times 5) \pmod{26} = 1$
- $P = 22 \Rightarrow (22 \times 5) \pmod{26} = 6$
- $P = 23 \Rightarrow (23 \times 5) \pmod{26} = 11$
- $P = 24 \Rightarrow (24 \times 5) \pmod{26} = 16$
- $P = 25 \Rightarrow (25 \times 5) \pmod{26} = 21$

4. Do different plaintext values ever produce the exact same ciphertext?

No. Every plaintext $P \in \{1, \dots, 25\}$ produces a unique ciphertext in $\{1, \dots, 25\}$.

5. If two distinct plaintexts produce the same ciphertext, what problem does this create for decryption?

This is called a **collision**. If two different plaintexts map to the same ciphertext, the encryption function is non-invertible (not one-to-one). The receiver cannot uniquely determine which original plaintext was sent.

Exercise 5: How Do We Divide Modulo m ?

1. Why does $P = \frac{22}{5}$ not work directly in modular arithmetic over integers? Modular arithmetic operates strictly over the set of integers $Z_m = \{0, 1, \dots, m - 1\}$. Fractional values like $\frac{22}{5} = 4.4$ do not exist in modular arithmetic. Division must be replaced by multiplication by a modular multiplicative inverse.

2. Find an integer satisfying $5x \equiv 1 \pmod{26}$ by testing $5(1), 5(2), \dots \pmod{26}$

- $5(1) = 5 \pmod{26} = 5$
- $5(2) = 10 \pmod{26} = 10$
- $5(3) = 15 \pmod{26} = 15$
- $5(4) = 20 \pmod{26} = 20$
- $5(5) = 25 \pmod{26} = 25$
- $5(6) = 30 \pmod{26} = 4$
- $5(7) = 35 \pmod{26} = 9$
- $5(8) = 40 \pmod{26} = 14$
- $5(9) = 45 \pmod{26} = 19$
- $5(10) = 50 \pmod{26} = 24$
- $5(11) = 55 \pmod{26} = 3$
- $5(12) = 60 \pmod{26} = 8$
- $5(13) = 65 \pmod{26} = 13$
- $5(14) = 70 \pmod{26} = 18$
- $5(15) = 75 \pmod{26} = 23$
- $5(16) = 80 \pmod{26} = 2$
- $5(17) = 85 \pmod{26} = 7$
- $5(18) = 90 \pmod{26} = 12$
- $5(19) = 95 \pmod{26} = 17$

- $5(20) = 100 \pmod{26} = 22$
- $5(21) = 105 \pmod{26} = 1$

3. State $5^{-1} \pmod{26}$

$$5^{-1} \pmod{26} = 21$$

4. Compute $P = (22 \times 5^{-1}) \pmod{26}$ to recover the plaintext

- $P = (22 \times 21) \pmod{26}$
- $22 \times 21 = 462$
- $462 \pmod{26} \Rightarrow 462 = (26 \times 17) + 20$
- $P = 20$

Exercise 6: Understanding the Multiplicative Inverse

1. Fill in Table 1 by calculating $\gcd(k, m)$

k	m	$\gcd(k, m)$	Inverse $k^{-1} \pmod{m}$ exists? (Yes/No)
5	26	1	Yes
7	26	1	Yes
2	26	2	No
13	26	13	No
3	26	1	Yes

k	m	gcd(k,m)	Inverse $k^{-1}(\text{mod } m)$ exists? (Yes/No)
5	25	5	No

2. For those that exist, find $k^{-1} \pmod{m}$ and verify $(k \times k^{-1}) \equiv 1 \pmod{m}$

- **Pair (5,26):** $k^{-1} = 21 \Rightarrow 5 \times 21 = 105 \equiv 1 \pmod{26}$
- **Pair (7,26):** $k^{-1} = 15 \Rightarrow 7 \times 15 = 105 \equiv 1 \pmod{26}$
- **Pair (3,26):** $k^{-1} = 9 \Rightarrow 3 \times 9 = 27 \equiv 1 \pmod{26}$

3. Complete the fundamental rule

$$k^{-1} \text{ exists modulo } m \Leftrightarrow \gcd(k, m) = 1$$

Exercise 7: Why Some Multiplication Keys Cannot Decrypt

Let $m = 26$ and $k = 2$. Encryption: $C = (2P) \pmod{26}$.

1. Calculate C for $P = 1, 2, 3, \dots, 13$

- $P = 1 \Rightarrow C = (2 \times 1) \pmod{26} = 2$
- $P = 2 \Rightarrow C = (2 \times 2) \pmod{26} = 4$
- $P = 3 \Rightarrow C = (2 \times 3) \pmod{26} = 6$
- $P = 4 \Rightarrow C = (2 \times 4) \pmod{26} = 8$
- $P = 5 \Rightarrow C = (2 \times 5) \pmod{26} = 10$
- $P = 6 \Rightarrow C = (2 \times 6) \pmod{26} = 12$
- $P = 7 \Rightarrow C = (2 \times 7) \pmod{26} = 14$
- $P = 8 \Rightarrow C = (2 \times 8) \pmod{26} = 16$
- $P = 9 \Rightarrow C = (2 \times 9) \pmod{26} = 18$
- $P = 10 \Rightarrow C = (2 \times 10) \pmod{26} = 20$
- $P = 11 \Rightarrow C = (2 \times 11) \pmod{26} = 22$
- $P = 12 \Rightarrow C = (2 \times 12) \pmod{26} = 24$
- $P = 13 \Rightarrow C = (2 \times 13) \pmod{26} = 0$

2. Show that $E(1) = E(14)$

- $E(1) = (2 \times 1) \pmod{26} = 2$
- $E(14) = (2 \times 14) \pmod{26} = 28 \pmod{26} = 2$
- Thus, $E(1) = E(14) = 2$.

3. If you receive ciphertext $C = 2$, can you uniquely determine whether the plaintext was 1 or 14?

No. Both $P = 1$ and $P = 14$ map to $C = 2$, so it is impossible to decide which plaintext was originally encrypted.

3. Can this encryption function be safely decrypted?

No, because it is not a one-to-one (bijective) function. Information is lost during encryption.

5. Compute $\gcd(2, 26)$ and explain how this relates to the collision problem

- $\gcd(2, 26) = 2 \neq 1$.
- Because $\gcd(k, m) > 1$, k share factors with m . This compresses the output space to only $26/2 = 13$ unique outputs, causing multiple plaintexts to collide into the same ciphertexts.

Exercise 8: Build a Complete Multiplication Cipher

2. Manually calculate $k^{-1} \pmod{26}$ for $k = 5$

Find x such that $5x \equiv 1 \pmod{26}$.

As shown in Exercise 5, $5 \times 21 = 105 = (26 \times 4) + 1$.

$$k^{-1} = 21$$

3. Encrypt $P = 20$, then decrypt using k^{-1}

- **Encryption:** $C = (20 \times 5) \pmod{26} = 100 \pmod{26} = 22$
- **Decryption:** $P_{\text{recovered}} = (22 \times 21) \pmod{26} = 462 \pmod{26} = 20$
- **Verification:** $P_{\text{recovered}} = 20 = P$, verifying correct execution.

Exercise 9: Key Generation

2. How many valid keys exist for $m = 26$?

The number of valid keys is given by Euler's Totient Function $\phi(26) = 26 \times \left(1 - \frac{1}{2}\right) \times \left(1 - \frac{1}{13}\right) = 12$. There are **12 valid keys**.

3. List the invalid key values. Why are they invalid?

- **Valid keys:** {1,3,5,7,9,11,15,17,19,21,23,25}
- **Invalid keys:** {2,4,6,8,10,12,13,14,16,18,20,22,24}
- **Reason:** They are invalid because $\gcd(k, 26) > 1$. They share a common factor with 26 (either 2 or 13), meaning they do not have a multiplicative inverse and cause collisions.

4. Compute the corresponding inverse $k^{-1} \pmod{26}$ for each valid k

k	Calculation	$k^{-1} \pmod{26}$
1	$1 \times 1 \equiv 1$	1
3	$3 \times 9 = 27 \equiv 1$	9
5	$5 \times 21 = 105 \equiv 1$	21
7	$7 \times 15 = 105 \equiv 1$	15
9	$9 \times 3 = 27 \equiv 1$	3
11	$11 \times 19 = 209 \equiv 1$	19
15	$15 \times 7 = 105 \equiv 1$	7

k	Calculation	$k^{-1} \pmod{26}$
17	$17 \times 23 = 391 \equiv 1$	23
19	$19 \times 11 = 209 \equiv 1$	11
21	$21 \times 5 = 105 \equiv 1$	5
23	$23 \times 17 = 391 \equiv 1$	17
25	$25 \times 25 = 625 \equiv 1$	25

Exercise 10: Sender and Receiver Scenario

1. Is $k = 7$ a valid multiplication key modulo 26?

Yes, because $\gcd(7, 26) = 1$.

2. Calculate $k^{-1} \pmod{26}$

$$7 \times x \equiv 1 \pmod{26} \Rightarrow 7 \times 15 = 105 \equiv 1 \pmod{26}.$$

$$k^{-1} = 15$$

3. Which key does Alice use to send messages, and which key does Bob use to read them?

- **Alice (Sender):** Uses encryption key $k = 7$.
- **Bob (Receiver):** Uses decryption key $k^{-1} = 15$.

4. Alice encrypts $P = 19$. Compute ciphertext C

- $C = (19 \times 7) \pmod{26}$
- $19 \times 7 = 133$
- $133 \pmod{26} \Rightarrow 133 = (26 \times 5) + 3$
- $C = 3$

5. Bob receives $C = 3$. Perform decryption calculation

- $P = (3 \times 15) \pmod{26}$
- $3 \times 15 = 45$
- $45 \pmod{26} \Rightarrow 45 = (26 \times 1) + 19$
- $P = 19$

6. Conceptual Question

Bob's key is not an unrelated random number; it is mathematically derived directly from Alice's key k and the public modulus m via the modular inverse algorithm.

Exercise 11: What Information Must Be Shared?

Given $m = 26$, encryption algorithm $C = (P \times k) \pmod{m}$, and secret key $k = 5$:

1. Which information can be made public without breaking the system?

- The modulus $m = 26$.
- The encryption algorithm structure $C = (P \times k) \pmod{m}$.

2. Which information must remain secret?

The key $k = 5$ (and by extension $k^{-1} = 21$).

3. If a receiver knows m and k , can they independently calculate k^{-1} ?

Yes. Calculating k^{-1} from k and m requires only the Extended Euclidean Algorithm, which is computationally trivial for a known k .

4. Does the sender need to explicitly transmit k^{-1} over the network?

No. As long as the receiver knows k and m , they can compute k^{-1} locally on their own device.

Exercise 12: Comparing Addition and Multiplication Ciphers

Property	Scheme A (Addition)	Scheme B (Multiplication)
Encryption Formula	$C = (P + 7) \pmod{26}$	$C = (P \times 7) \pmod{26}$
Decryption Formula	$P = (C - 7) \pmod{26}$	$P = (C \times 15) \pmod{26}$
Inverse Operation Required	Additive Inverse ($-k$ or $m - k$)	Multiplicative Inverse ($k^{-1} \pmod{m}$)
Is Every Key Reversible?	Yes (all integers $k \in \mathbb{Z}_m$ work)	No
Reversibility Condition	None (always reversible)	$\gcd(k, m) = 1$

Exercise 13: Introduction to Modular Exponentiation

1. Compute $3^4 \pmod{26}$ by hand

- $3^4 = 81$
- $81 \pmod{26} \Rightarrow 81 = (26 \times 3) + 3$
- $C = 3$

3. Compute C for $P \in \{1, 2, \dots, 25\}$ using $k = 4$

Formula: $C = P^4 \pmod{26}$

- $P = 1 \Rightarrow 1^4 = 1 \pmod{26} = 1$
- $P = 2 \Rightarrow 2^4 = 16 \pmod{26} = 16$
- $P = 3 \Rightarrow 3^4 = 81 \pmod{26} = 3$
- $P = 4 \Rightarrow 4^4 = 256 \pmod{26} = 22$

- $P = 5 \Rightarrow 5^4 = 625 \pmod{26} = 1$
- $P = 6 \Rightarrow 6^4 = 1296 \pmod{26} = 22$
- $P = 7 \Rightarrow 7^4 = 2401 \pmod{26} = 9$
- $P = 8 \Rightarrow 8^4 = 4096 \pmod{26} = 14$
- $P = 9 \Rightarrow 9^4 = 6561 \pmod{26} = 9$
- $P = 10 \Rightarrow 10^4 = 10000 \pmod{26} = 16$
- $P = 11 \Rightarrow 11^4 = 14641 \pmod{26} = 3$
- $P = 12 \Rightarrow 12^4 = 20736 \pmod{26} = 14$
- $P = 13 \Rightarrow 13^4 = 28561 \pmod{26} = 13$
- $P = 14 \Rightarrow 14^4 = 38416 \pmod{26} = 14$
- $P = 15 \Rightarrow 15^4 = 50625 \pmod{26} = 3$
- $P = 16 \Rightarrow 16^4 = 65536 \pmod{26} = 16$
- $P = 17 \Rightarrow 17^4 = 83521 \pmod{26} = 9$
- $P = 18 \Rightarrow 18^4 = 104976 \pmod{26} = 14$
- $P = 19 \Rightarrow 19^4 = 130321 \pmod{26} = 9$
- $P = 20 \Rightarrow 20^4 = 160000 \pmod{26} = 22$
- $P = 21 \Rightarrow 21^4 = 194481 \pmod{26} = 1$
- $P = 22 \Rightarrow 22^4 = 234256 \pmod{26} = 22$
- $P = 23 \Rightarrow 23^4 = 279841 \pmod{26} = 3$
- $P = 24 \Rightarrow 24^4 = 331776 \pmod{26} = 16$
- $P = 25 \Rightarrow 25^4 = 390625 \pmod{26} = 1$

4. Is reversing modular exponentiation as straightforward as subtracting k or multiplying by an inverse?

No. Reversing modular exponentiation requires solving for an exponent d such that $(P^e)^d \equiv P \pmod{m}$, which relies on Euler's totient function $\phi(m)$, or solving discrete logarithms, which is mathematically much harder.

5. What mathematical problem must be solved to recover P from $P^k \equiv C \pmod{m}$?

The **Discrete Logarithm Problem (DLP)** or finding the modular exponent inverse using Euler's Totient Function $\phi(m)$.

Exercise 14: Comprehensive Final Challenge

Part 1: Analytical Steps ($m = 29$, $k = 12$, $P = 17$)

A. Key Validation

Compute $\gcd(12, 29)$. Since 29 is prime and does not divide 12, $\gcd(12, 29) = 1$.

Result: $k = 12$ is a **valid key**.

B. Complementary Key

Find $12^{-1} \pmod{29}$ such that $12x \equiv 1 \pmod{29}$:

- $12 \times 1 = 12$
- $12 \times 2 = 24$
- $12 \times 3 = 36 \equiv 7 \pmod{29}$
- $12 \times 4 = 48 \equiv 19 \pmod{29}$
- $12 \times 5 = 60 \equiv 2 \pmod{29}$
- Since $12 \times 5 \equiv 2 \pmod{29}$, multiply by 15:
 $12 \times (5 \times 15) = 12 \times 75 \equiv 2 \times 15 = 30 \equiv 1 \pmod{29}$.
- Reduce 75 modulo 29: $75 = (29 \times 2) + 17$.
- **Check:** $12 \times 17 = 204 = (29 \times 7) + 1 = 203 + 1 \equiv 1 \pmod{29}$.

$$12^{-1} \pmod{29} = 17$$

C. Encryption

$$C = (17 \times 12) \pmod{29}$$

- $17 \times 12 = 204$
- $204 \pmod{29} \Rightarrow 204 = (29 \times 7) + 1$

$$C = 1$$

$$\text{D. Decryption } P = (C \times 12^{-1}) \pmod{29} = (1 \times 17) \pmod{29}$$

$$P = 17$$

E. Verification

$$P_{\text{original}} = 17 = P_{\text{decrypted}}$$

F. Conceptual Questions

i. Why does Bob specifically require 12^{-1} ?

Because 12^{-1} is the unique multiplicative inverse that satisfies $(12 \times 12^{-1}) \equiv 1 \pmod{29}$, undoing the multiplication performed during encryption.

ii. Why isn't the modulus $m = 29$ kept secret? The modulus m defines the algebraic system (field bounds) and is part of the public parameters. System security relies entirely on keeping the key k private, not the modulus.

iii. What would happen if Alice had chosen $k = 2$?

Since $\gcd(2, 29) = 1$ (29 is prime), $k = 2$ would still be a valid key with inverse $2^{-1} = 15 \pmod{29}$.

iv. Does Bob need k^{-1} sent to him directly?

No. Bob can calculate k^{-1} independently using $k = 12$ and $m = 29$ via the Extended Euclidean Algorithm.